

ME 620: Modeling and Simulation

Course Code	ME 620
Course Name	Modeling and Simulation
L-T-P-C	2-0-2-3
Prerequisites	Engineering Mathematics (IC110), Linear Algebra (IC111)
Intended for	UG/MS/PhD
Distribution	Elective
Approval	6th Senate (May 2014)

Course Contents

- **Introduction**
System, environment, input and output variables, State variables; Static and Dynamic systems; Hierarchy of knowledge about a system and Modeling Strategy. [2 Lectures]
- **Physical Modeling**
Dimensions analysis, Dimensionless grouping of input and output variables of find empirical relations, similarity criteria and their application to physical models. [2 Lectures]
- **Modeling of System with Known Structure**
Review of conservation laws and the governing equation for heat, mass and momentum transfer, Deterministic model- (a) distributed parameter models in terms of partial identification and their solutions and (b) lumped parameter models in terms of differential and difference equations, state space model, transfer functions block diagram and sub systems, stability of transfer functions, modeling for control. [7 Lectures]
- **Optimizations and Design of Systems**
Summary of gradient based techniques : Nontraditional Optimizations techniques (1) genetic Algorithm (GA)- coding, GA operations elitism, Application using MATLAB:(ii) Simulated Annealing. [4 Lectures]
- **Neural Network Modeling of Systems only with Input-output Database**
Neurons, architecture of neural networks, knowledge representation, learning algorithm. Multilayer feed forward network and its back propagation learning algorithm, Application to complex engineering systems and strategy for optimum output. [4 Lectures]
- **Modeling Based on Expert Knowledge**
Fuzzy sets, Membership functions, Fuzzy Inference systems, Expert Knowledge and Fuzzy Models, Design of Fuzzy Controllers. [4 Lectures]
- **Simulation of Engineering Systems**
Monte-Carlo simulation, Simulation of continuous and discrete processes with suitable examples from engineering problems. [5 Lectures]

List of practicals

- Exp. 1 & 2 Introductions to programming with MATLAB
- Exp. 3 & 4 Find the response of a lumped variable model expressed in terms of transfer function using MATLAB for input of (i) unit step function (ii) unit impact function and (iii) unit ramp function
- Exp. 5, 6 & 7 Use of Simulink in MATLAB for engineering problems
- Exp. 8, 9 & 10 Use of Neural Network in MATLAB for engineering problems
- Exp. 10, 11 & 12 Use of FIS and ANFIS in MATLAB for engineering problems
- Exp. 13 & 14 Monte Carlo simulation

Text Books

1. Zeigler B.P., Praehofer. H. and Kim I.G. **Theory of modeling and simulation**, 2nd Edition, Academic press, 2000
2. Ogata K, **Modern control Engineering**, 3rd Edition, Prentice hall of India, 2001
3. Jang J.S.R., sun C.T. and Mizutani E., **Neuro-Fuzzy and soft Computing**, 3 rd Edition, Prentice hall of India, 2002.
4. Shannon, R. E., **System Simulation: the Art and Science**, Prentice Hall Inc., 1990.
5. Pratap R., **Getting started with MATLAB**, Oxford university Press 2009